Project overview

Context and objectives

This project supports a clinical data science workflow using the **colorectal_cancer_prediction_cleaned_imputed.csv** dataset to explore prognostic factors in colorectal cancer and build interpretable prediction/survival models. The goal is to produce reproducible analyses, transparent methodology, and publication-ready figures/tables.

Clinical questions

- **Primary:** Which clinical and pathological features are associated with colorectal cancer outcomes?
- **Secondary:** How do model-based risk scores compare to traditional staging (e.g., TNM, tumor grade)?
- **Exploratory:** Can interpretable feature effects (coefficients, SHAP, partial dependence) improve clinical understanding?

Dataset summary

- **Source:** colorectal_cancer_prediction (cleaned + imputed version)
- **File:** colorectal_cancer_prediction_cleaned_imputed.csv
- **Key variables:** survival time, event indicator, age, tumor size, nodal status, grade, biomarkers, treatment indicators
- **Missingness handling:** Imputation performed during preprocessing; details documented in docs/methods.md

Reproducible workflow

- **Environment:** Google Colab + pinned Python versions
- **Pipeline:** Numbered notebook sections (00–99) for setup, EDA, preprocessing, modeling, evaluation, reporting
- **Outputs:** All figures/tables saved under outputs/ with traceable file stems
- **Versioning:** GitHub + commit messages describing changes to data processing and modeling

Deliverables

- **EDA figures:** Histograms, correlation heatmaps, Kaplan–Meier curves
- **Model artifacts:** Trained Cox PH / RSF models + serialized coefficients/feature importances
- **Reports:** Markdown summaries under docs/; metrics under outputs/metrics/
- **Interpretability:** Coefficient tables, partial dependence, ICE, SHAP

Directory map

- **docs/** Project overview, methodology, references
- **notebooks/** Numbered Colab notebooks
- **outputs/** Saved plots, tables, metrics
- **data/** colorectal_cancer_prediction_cleaned_imputed.csv
- **src/** Reusable preprocessing/modeling functions

Quick start

1. Open Colab: notebooks/00_setup_environment.ipynb
2. Load data: data/colorectal_cancer_prediction_cleaned_imputed.csv
3. Run EDA: notebooks/10_eda_profile.ipynb
4. Preprocess + encode: notebooks/20_preprocess_encode.ipynb
5. Train models: notebooks/30_modeling.ipynb
6. Generate report: notebooks/90_reporting.ipynb